

Software Test Plan

Project Title: School Management System



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STTPK3023 SOFTWARE TESTING (A)

SOFTWARE TEST PLAN: SCHOOL MANAGEMENT SYSTEM

SUBMITTED TO:

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Purpose of the Document

The purpose of this Test Plan is to define the testing strategy and overall testing framework for the School Management System. This system states the scope of testing, testing strategy, execution strategies, test data approach, and roles and responsibilities involved in the testing process. This Test Plan will act as a guide to ensure the system meets its functional requirements and is ready for user acceptance.

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1 Introduction

1.1 Purpose

The School Management System (SMS) is a web-based application, developed to support the academic and administrative activities in a school environment. This system allows users such as administrators, teachers, students, and system owners to view their respective records and relevant data through their assigned platform.

This test plan acts as a guide to ensure every module is executed correctly, through the assistance of automated tools, to detect errors early, reduce manual testing process, and improve the system quality as a whole.

1.2 Project Overview

This project involves the testing of a School Management System developed as an academic software project. The system is intended for use by school administrators and teachers to manage daily academic operations such as student enrollment, class management, subject allocation, and attendance tracking.

The version of the system under test includes core functional features without integration to external third-party services. Testing will be conducted in a controlled testing environment using dummy data to simulate real-world usage scenarios. The focus of this project is to validate functional correctness, usability, and workflow accuracy, instead of performance optimization or security testing.

The purpose of the School Management System are:

- 1) To provide a centralized platform for managing core academic and administrative operations, including student records, teacher records, attendance, timetables, and examination results.
- 2) To streamline administrative tasks such as leave management, notice publishing, syllabus and notes uploading.
- 3) To support effective communication and information sharing between administrators and teachers through centralized content and announcement management.
- 4) To enhance system accessibility and usability through features such as single login authentication, password recovery, multi-language support, and dark theme support.

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2 Scope

2.1 In-Scope

The automated system used for testing, which is Playwright, will be used to cover the main features in the School Management System (SMS).

2.1.1 Functional Scope

Feature	Description
User Authentication	Login for admin, teacher, student, and owner, where it covers input validation.
Teacher Management	Manage teachers, show a list of teachers registered in school, and manage leave applications.
Student Management	Manage students, display list of students (according to class and section), view feedback for students.
Subject Monitoring	Manage subjects for each class.
Attendance List	View attendance for students according to each class and section.
Notice Board	Create notice and view notice board.
Time Table	View time table for each class and section.
Syllabus	View and upload syllabus for each class.
Notes	View notes for each class.
Marks	Upload and view marks for all students.
Bus Service	Manage bus services and bus drivers.
Settings	Edit profile (restricted to admin), change password (available for all).
Apply Leave	Apply leave, modify, delete and view list of applied leaves made.

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2.1.2 Interface Scope

The testing of this system includes:

- All relevant system pages and user interface components for Admin and Teacher dashboards
- Functional verification of buttons, input forms, pop-ups, and navigation menus across tested modules
- Layout behavior and page consistency during navigation between system modules
- Validation of error messages displayed for invalid inputs and failed operations
- Verification of confirmation messages for successful actions such as record creation, updates, and submissions

Using Playwright as an automated tool ensures the key user interface responds to user interactions and functions consistently across the tested modules of the system.

2.1.3 Workflow Scope

The workflow scope covers for main users, which are admin and teacher. The workflow for admin and teacher are mostly similar, with one key difference of teacher, is that teacher can apply for leaves.

Admin Flow:

1. Admin Login: The administrator enters a registered email and password. The system validates the credentials and displays the admin dashboard upon successful authentication.
2. Manage Teachers: The administrator views the list of registered teachers, adds new teacher records, updates teacher information, and reviews teacher leave applications.
3. Manage Students: The administrator accesses student records, views student lists based on class and section, and reviews feedback related to students.
4. Manage Subjects: The administrator creates and manages subjects assigned to different classes to ensure accurate subject monitoring.

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5. View Attendance: The administrator views attendance records for students according to their respective classes and sections.
6. Manage Notice Board: The administrator creates and publishes notices and views the notice board for school-wide communication.
7. Manage Timetable: The administrator views timetables assigned to each class and section to support academic scheduling.
8. Manage Syllabus: The administrator uploads and updates syllabus files for each class to ensure students have access to learning materials.
9. Manage Notes: The administrator uploads and manages notes for each class and subject to support learning activities.
10. Manage Examination Marks: The administrator uploads examination marks and views academic performance records for all students.
11. Manage Bus Services: The administrator manages bus services, including bus and driver information, to support student transportation.
12. Manage Account Settings: The administrator updates profile information and manages password changes for account security.

Teacher Flow:

1. Teacher Login: The teacher enters a registered email and password. The system validates the credentials and displays the teacher dashboard upon successful authentication.
2. View Student List: The teacher accesses the list of students according to assigned classes and sections for academic reference.
3. Manage Attendance: The teacher monitors attendance records for students in their assigned classes and sections.
4. Create and View Notices: The teacher creates notices and views notices published by the administrator through the notice board.

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5. View Time Table: The teacher views the time table for each class, assigned by the administrator.
6. Manage Syllabus: The teacher accesses syllabus files uploaded for each assigned class and subject.
7. Manage marks: The teacher uploads and views marks for students, according to the quiz made.
8. Apply Leave: The teacher submits a leave application by entering leave details such as leave type, description, start date, and end date. The system validates the input and displays the submitted leave in the leave list.
9. Modify or Delete Leave Application: The teacher edits or deletes a previously submitted leave application before approval to ensure accurate leave records.
10. Manage Profile: The teacher updates personal profile picture and changes password for security purposes.

2.2 Out-of-Scope

The features excluded from the testing are as follows:

Features	Reasons
Performance and Load Testing	Performance and load testing were excluded from this test scope, as the objective of this testing was to validate functional correctness rather than system scalability.
Advanced Security Testing	Advanced security testing was not included in this test scope, where it is only valid to basic authentication and access control verification, as they require specialized security expertise.
Testing for Other User Roles	Testing for Student and Owner roles was excluded from this report. The scope of testing focused solely on Admin and Teacher roles, as these roles are involved in the core administrative and management functionalities evaluated.

3 Testing Strategy

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3.1 Test Objectives

- 1) To verify login functionality and role-based dashboard navigation for Admin and Teacher users using page navigation assertions.
- 2) To validate core administrative functions, including teacher management, student management, subject management, and record viewing, through form validation and UI interaction checks.
- 3) To ensure content management features such as notice creation, syllabus upload, and notes upload function correctly by verifying successful file uploads and confirmation messages.
- 4) To identify, record, and track defects discovered during functional and automated testing.

3.2 Test Assumptions

The testing process assumes that:

- Playwright is installed, properly configured, and functioning correctly.
- Testers have access to the deployed system and required test environments.
- Admin and Teacher test accounts are prepared and available for testing.
- The system built is stable to support repeated functional and automated test execution.

3.3 Data Approach

1. Prepare Admin and Teacher accounts with appropriate role permissions.
2. Create test data for teachers, students, subjects, notices, syllabus files, and notes.
3. Use dummy leave applications to test leave submission, validation, modification, and deletion scenarios.
4. Apply consistent and reusable test data across test cycles to ensure reliable results.

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3.4 Level of Testing

Test Type	Description	Responsible Parties
Unit Testing	Testing individual system components and functions such as login validation, input validation for forms, and password reset functionality to ensure each unit works correctly in isolation.	Siti Ain Athiqah (Test Lead)
Functional Testing	Testing core system features to verify they operate according to requirements, such as teacher and student management, subject management, and role-based dashboard navigation for Admin and Teacher users.	Hana Syakirah (Test Analyst)
User Acceptance Testing	Ensuring that the system meets user requirements by validating real-world usage flows. This testing focuses on system usability, correctness of workflows, and overall user experience.	Muhammad Haziq (Test Analyst)

3.5 Unit Testing

Unit testing is conducted to verify that individual system components or modules function correctly in isolation. Each unit is tested independently using predefined test cases to ensure that the input, processing logic, and output meet the specified requirements. Unit testing is mainly performed during the development phase to detect errors early.

Testing focuses on small units such as functions, classes, forms, or validation logic without involving other system components.

Features to be Tested:

- Page navigation assertions
- Text field or form validation checks
- Error message handling
- File upload verification

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Participants:

Tester's Name	Department/ Area	Role
Dr Mawarny Binti Md Rejab	SOC, UUM	Test Manager
Siti Ain Athiqah Binti Sahrn	SOC, UUM	Test Lead
Hana Syakirah Binti Hassan Khairullah	SOC, UUM	Test Analyst
Muhammad Haziq Bin Idrus	SOC, UUM	Test Analyst

3.6 Functional Testing

Functional testing is carried out to ensure that each system function works according to the specified user requirements. The testing is performed by executing system features through the user interface and validating system behavior against expected outcomes.

Features to be Tested:

- User authentication (Login)
- Profile and password management
- Student management
- Teacher management
- Subject and syllabus
- Notice board
- Notes
- Data validation and error handling messages

Testing Approach

- Both positive and negative scenarios are tested
- System responses are compared with expected results
- Defects are logged and reported for correction

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Participants:

Tester's Name	Department/ Area	Role
Dr Mawarny Binti Md Rejab	SOC, UUM	Test Manager
Siti Ain Athiqah Binti Sahrn	SOC, UUM	Test Lead
Hana Syakirah Binti Hassan Khairullah	SOC, UUM	Test Analyst
Muhammad Haziq Bin Idrus	SOC, UUM	Test Analyst

3.7 User Acceptance Testing

User Acceptance Testing (UAT) is conducted to ensure that the system meets user expectations and business requirements. The testing is performed by the end user or system owner in a real-world scenario to confirm that the system is ready for deployment.

Participants:

Tester's Name	Department/ Area	Role
Dr Mawarny Binti Md Rejab	SOC, UUM	Test Manager

4 Execution Strategy

4.1 Entry Criteria

- The entry criteria refer to the desirable conditions in order to start test execution
- Entry criteria are flexible benchmarks. If they are not met, the test team will assess the risk, identify mitigation actions and provide a recommendation.

Entry Criteria	Test Team	Technical Team	Notes
Test environment(s) is available			The environment is stable and accessible to the team
Test data is available			Representative test data prepared

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Code has been merged successfully			No merge conflicts or build errors
Development has completed unit testing			No critical defect
Playwright test scripts are completed, reviewed and approved by the Project Team			Cases reviewed

4.2 Exit Criteria

- o The exit criteria are the desirable conditions that need to be met in order to proceed with the implementation.
- o Exit criteria are flexible benchmarks. If they are not met, the test team will assess the risk, identify mitigation actions, and provide a recommendation.

Exit Criteria	Test Team	Technical Team	Notes
100% Test Scripts executed			
90% pass rate of Test Scripts			
No open Critical and High severity defects			
All remaining defects are either cancelled or documented as Change Requests for a future release			
All expected and actual results are captured and documented with the test script			
All test metrics collected based on reports from daily and Weekly Status reports			
All defects logged in Defect Tracker/Spreadsheet			
Test environment cleanup completed and a new back up of the environment			

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4.3 Validation and Defect Management

4.3.1 Validation of Test Cases / Test Scenarios

The test cases and scenarios will be run through an automated testing tool called Playwright with the test scripts. All test cases will be based on the system's features and requirements. Aside from that, manual testing will be performed to ensure the UI usability and responsiveness.

4.3.2 Defect Management Process

The testers will execute all approved Playwright test scripts for each feature or requirement. Playwright will automatically run tests across browsers and environments as required. Any test that fails in Playwright indicates a potential defect in the system. The playwright records failure information in a report that includes error messages, screenshots of the failure component, and video recordings of the test run. Testers will use the report as evidence to log defects, which will then be fixed by the technical team.

- Playwright will automatically validate expected and actual results through:
 - o Page navigation assertions
 - o Text field or form validation checks
 - o File upload verification
- Defects found during the testing should be categorized as below:

Severity	Impact
1 (Critical)	▪ System functionality is completely blocked. Users cannot perform essential tasks.
2 (High)	▪ Major functionality is broken. Users cannot perform key tasks, but testing of other unrelated modules can continue.
3 (Medium)	▪ Functionality has issues, but a workaround exists. Users can still complete tasks with extra effort.
4 (Low)	▪ Minor issues that do not affect core functionality. Cosmetic or usability issues.

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5 Environment Requirements

5.1 Test Environments

- Environment(s) requirements

Category	Software / Tool	Version / Configuration
Operating System	Windows	Windows 10 or 11 (64-bit)
Processor	Intel Core	i5 Processor and above
RAM	-	16 GB
Server	XAMPP	Apache + MySQL (must support PHP 8.1)
Browser	Google Chrome	Latest version
Automation Tool	Playwright	Latest version
Runtime Environment	Node.js	LTS Version
IDE	Visual Studio Code	Latest version with PlayWright Extension installed

- Security requirements
 - **User Role Configuration:** User is restricted to only access their defined roles and interface.
 - **Test Data Privacy:** Test environments do not use any real student or teacher data.
 - **Use of Default Password:** The environment testing will use default email and password.
 - **Session Handling:** Session token is destroyed immediately after logout and does not allow re-entry via Back button.

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6 Significantly Impacted Division/College/Department

Business Area	Business Manager	Tester(s)
Administration	School Administrator	Ain, Haziq
Academic Staff	Head Teacher	Hana
Student Affairs	Student Representative	Ain, Hana
IT Support	System Owner	Haziq

7 Dependencies

- **Code Stability:** Ensure all SMS modules that undergo testing should be completed and merged into the main branch, and ensure that no code changes during testing.
- **Server Availability:** Ensure that XAMPP is successfully running with Apache (Port 80) and MySQL (Port 3306) without port conflicts.
- **Database Synchronization:** Ensure that the database schema (*_sms.sql*) successfully imported before running the automation scripts.
- **Automation Tool Setup:** Ensure that Node.js and PlayWright dependencies are configured correctly and the browsers are downloaded.
- **Test Data Availability:** Ensure that valid credentials for SMS are created and verified before the test execution.